

Module 04: Cognition in Embodied

Learning Objectives

1. Define **Cognition** within the context of Embodied.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between Perception and Action.

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In Embodied, we see Cognition manifest in: * **Specific Example 1:** When I walk through an unfamiliar city and build a mental map of the streets, my spatial cognition is grounded in the embodied experience of my own locomotion -- I remember the route not as an abstract diagram but as a sequence of bodily turns, uphill efforts, and landmark encounters; this is the generative model encoding spatial knowledge through proprioceptive and vestibular predictions, so that "knowing where I am" is inseparable from the body's felt history of moving through the space. * **Specific Example 2:** A basketball point guard surveying the court processes the positions and velocities of ten players not as a computational geometry problem but as an embodied spatial cognition that feels the open passing lane as a bodily impulse in the throwing arm and shoulder; the generative model integrates visual perception of spatial relationships with motor predictions of the pass trajectory, producing a form of movement cognition where thinking and doing are fused in a single predictive loop that minimizes free energy across spatial and motor domains simultaneously.

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.